Ansil M B

+91 8547 826 147 — ansiljbs@gmail.com — linkedin.com/in/ansilmbabl — github.com/ansilmbabl — medium.com/@ansilproabl

AI/ML Engineer and Data Scientist with 3+ years of experience building production-oriented Generative AI, computer vision, and machine learning solutions. Strong in Python, LLM applications, RAG, agentic workflows, model integration, and AI-backed automation. Experienced in taking solutions from data and prototype through APIs, containerized deployment, testing, tracing, and production support. Targeting AI Engineer, GenAI Engineer, Applied ML, Computer Vision, and AI Backend roles.

Experience

Grid Dynamics Pvt. Ltd. **Sep 2025 – Present**
Data Scientist — Contingent worker supporting Visa *Bengaluru, India*

Management Information System — MCP-enabled Business Analyst Agent

- Built and deployed a Business Analyst agent for an internal management information system in the financial-services domain using Python, JavaScript/Node.js, LLM APIs, and enterprise data sources; the system comprised 5 agents and 14 tools.
- Supported an internal AI application serving approximately 500 users and processing about 50 GB of data; reduced a reporting workflow from roughly two months of manual work to two weeks.
- Exposed the complete application as an MCP tool/server so other applications could consume its capabilities directly without additional integration hurdles.
- Developed RAG pipelines with document ingestion, chunking, embeddings, semantic and hybrid search, reranking, metadata filtering, query rewriting, pgvector, retrieval evaluation, and access-controlled retrieval.
- Integrated OpenAI, Anthropic, and Gemini models using LangChain, LangGraph, and MCP; implemented hallucination checks, DeepEval, latency tracking, prompt/version comparison, token and cost tracking, caching, and prompt optimization.
- Implemented backend services and data integrations using async Python, FastAPI, Pydantic, PostgreSQL, AWS S3, REST/gRPC APIs, microservices, OpenAPI, Docker, CI/CD, and on-premise Kubernetes on Red Hat clusters.
- Implemented security and reliability controls including AD group management, SSH key management, firewalls, JWT, HTTP bearer tokens, secrets management, Valkey caching, retries, fallbacks, logging, and error handling.
- Contributed across the software delivery lifecycle with Pytest, unit and integration testing, type hints, code reviews, production support, LangSmith tracing, operational dashboards, alerts, and live failure tracking.
- Worked directly with business stakeholders to translate requirements into agent and tool workflows, gathered feedback from internal users, and iterated on the application based on observed needs.
- Coordinated a four-person delivery team while owning hands-on development and technical problem-solving for assigned workstreams.
- Collaborated on frontend integration, including React-based interfaces, to expose AI-assisted workflows to internal users.

Aptagrim Consulting Ltd **Oct 2023 – Jul 2025**
AI Engineer

- **Pharmaceutical Research:** Built and integrated GenAI assistants using LLMs, vector databases, RAG, and agentic workflows; supported patient recruitment, drug-demand, and inventory forecasting and reduced human effort and analysis time by 50%.
- **Survey Automation and Validation:** Fine-tuned open- and closed-source LLMs to automate programming from unstructured requirements to XML and Excel outputs, reducing turnaround time from one day to under five minutes; achieved approximately 80% accuracy with open models and over 90% with closed-source models. Designed reviewer and inter-agent workflows with hallucination checks, prompt/version comparison, latency tracking, and token/cost tracking.
- **Computer Vision:** Developed and integrated oil-and-gas pressure chart reading systems using OpenCV and keypoint detection, achieving over 95% accuracy with processing time below three seconds; integrated results into client software to eliminate manual rounding errors and reduce data entry.
- **Automated Card Grading:** Built a system combining segmentation, object detection, OCR, and LLMs; developed custom grading models that were 80% smaller and more accurate than traditional models, reducing assessment time from approximately five minutes to seconds and enabling mobile image-based grading.
- **HR Assistant:** Developed and deployed an internal assistant using LLMs, async Python, FastAPI, Pydantic, SQL, structured outputs, retries, fallbacks, document retrieval, role-based access control (RBAC), and workflow automation, reducing HR support queries by 40%.
- Coordinated a company AI hackathon in 2024, leading planning, mentoring, and knowledge-sharing activities to support talent development.

Skills

- **Programming & Backend:** Python, async Python, JavaScript, Node.js, SQL, PostgreSQL, Bash, REST APIs, gRPC, FastAPI, Flask, microservices, OpenAPI
- **Machine Learning:** PyTorch, TensorFlow, ONNX, scikit-learn, Pandas, NumPy, model inference and optimization
- **Generative AI:** LLM applications, RAG, document ingestion, chunking, embeddings, semantic search, hybrid search, reranking, metadata filtering, query rewriting, prompt engineering, fine-tuning, structured outputs
- **Agents & Evaluation:** Agentic workflows, tool calling, LangChain, LangGraph, MCP, LangSmith tracing, DeepEval, hallucination checks, retrieval evaluation, latency tracking, prompt/version comparison, token and cost tracking, guardrails, multi-agent communication
- **Computer Vision:** OpenCV, scikit-image, Pillow, Albumentations, YOLO, SAM, Tesseract, OCR, segmentation, object detection
- **Data & Storage:** PostgreSQL, pgvector, MySQL, MongoDB, Redis, Valkey, FAISS, Pinecone, Weaviate, Chroma, Milvus, AWS S3
- **Deployment & MLOps:** Docker, on-premise Kubernetes, Red Hat clusters, CI/CD, Git, GitHub Actions, GitLab CI, Prometheus, Grafana, ONNX Runtime
- **Security & Reliability:** AD group management, SSH key management, firewalls, JWT, HTTP bearer tokens, secrets management, retries, fallbacks, logging, error handling, Pytest, unit testing, integration testing, type hints, code reviews
- **Frontend:** React, Streamlit

Education

IIIT Kottayam **2025 – Present**
Master of Technology in AI, ML and Data Science

IIT Palakkad **2022 - 2023**
Diploma in Artificial Intelligence and Machine Learning

APJ Abdul Kalam Technological University **2018 - 2022**
Bachelor of Technology in Mechanical Engineering
CGPA: 8.02

Certifications

- Machine Learning Specialization - Stanford University
- Deep Learning Specialization - DeepLearning AI
- Project Management Specialization - Google